PROBLEM DA9

Power of an Airplane Propeller

Original statement · DA9.pdf

Π theorem | Reynolds and Mach | Standard atmosphere

OFFICIAL PROBLEM STATEMENT

The power W needed to move the propeller of a small airplane depends on the airplane velocity V, propeller diameter D, angular velocity Ω, and also on the air properties, including its density ρ, viscosity μ, and sound velocity c; the latter, characterizing compressibility effects, has dimensions $[c] = LT^{-1}$. Assuming that the effect of gravity is negligible:

1. Use dimensional analysis to express the propeller power W as a function of the minimum number of dimensionless parameters.

A new airplane with propeller of diameter $D = 1.5$ m is to be designed to fly at an altitude $z = 5000$ m with cruise velocity $V = 300$ Km/h and propeller angular velocity $\Omega = 2500$ rpm. To determine the needed power, an experimental model is to be tested in a wind tunnel at sea level ($z = 0$ m).

2. Determine the model diameter D_e, the propeller angular velocity Ω_e, and the wind-tunnel velocity V_e needed to guarantee that the experiment provides a physically similar representation of the airplane flight conditions.

3. For the values of D_e, Ω_e, and V_e determined above, the power measured in the experiment (obtained as the product of Ω_e and the torque acting on the propeller shaft) is $W_e = 6 \times 10^7$ Watt. Obtain the power W corresponding to the real airplane.

Note: Physical properties for air according to the standard atmosphere:

z (m)	T (K)	ρ (kg/m^3)	μ (kg/(m s))	c (m/s)
0	288.15	1.225	1.775×10^{-5}	340.3
5000	255.65	0.736	1.630×10^{-5}	320.5

The value $D = 1.5$ m is written by hand on the printed sheet, next to "D".

Source: DA9.pdf (page 1, exam of Jan 10, 2012, statement and handwritten key). Notes.pdf, Chapter 6: Π theorem, Eq. 6.25; physical similarity, Eqs. 6.57-6.59.

Worked solution

Hypotheses, Data and Dimensional Table

- Air is a compressible gas: its compressibility enters through the speed of sound c (Notes.pdf, p. 66). Gravity is negligible.
- Prototype at $z = 5000$ m: $\rho = 0.736$ kg/m^3, $\mu = 1.630 \times 10^{-5}$ kg/(m s), $c = 320.5$ m/s. Model at $z = 0$: $\rho_e = 1.225$ kg/m^3, $\mu_e = 1.775 \times 10^{-5}$ kg/(m s), $c_e = 340.3$ m/s.

Parameter	**Dimensions**
W (unknown)	ML^2T^{-3}
ρ	ML^{-3}
μ	$ML^{-1}T^{-1}$
c, V	LT^{-1}
D	L
Ω	T^{-1}

Application of the Π Theorem

$$W = W(\rho, \mu, c, V, D, \Omega)$$

$n = 6$; (ρ, V, D) have independent dimensions (det $= -1$, as in DA1), so $n - k = 3$ groups remain.

- **Power.** $[W] = [\rho][V]^3[D]^2$ (exponents as in DA1): $\Pi_0 = W/(\rho V^3 D^2)$.
- **Viscosity.** $[\mu] = [\rho][V][D]$: the Reynolds number $\rho V D/\mu$.
- **Speed of sound.** $[c] = [V]$: the inverse Mach number V/c is the group.
- **Rotation.** $[\Omega] = [V][D]^{-1}$: the group $\Omega D/V$ (the inverse of the advance ratio).

$$\frac{W}{\rho V^3 D^2} = f\left(\frac{\rho V D}{\mu}, \frac{V}{c}, \frac{\Omega D}{V}\right)$$

Physical Similarity at Sea Level

The three groups must be equal. The Mach number fixes the tunnel speed:

$$\frac{V}{c} = \frac{V_e}{c_e} \implies V_e = \frac{c_e}{c} V = \frac{340.3}{320.5} \times 300 = 1.0618 \times 300 = 318.5 \text{ km/h} \simeq 319 \text{ km/h}$$

The Reynolds number then fixes the diameter:

$$\frac{\rho V D}{\mu} = \frac{\rho_e V_e D_e}{\mu_e} \implies D_e = \frac{\mu_e}{\mu}\,\frac{\rho}{\rho_e}\,\frac{V}{V_e}\,D = \frac{\mu_e}{\mu}\,\frac{\rho}{\rho_e}\,\frac{c}{c_e}\,D$$

$$D_e = \frac{1.775}{1.630} \times \frac{0.736}{1.225} \times \frac{320.5}{340.3} \times 1.5 = 1.0890 \times 0.6008 \times 0.9418 \times 1.5 = 0.924 \text{ m}$$

The advance ratio fixes the rotation:

$$\frac{\Omega D}{V} = \frac{\Omega_e D_e}{V_e} \implies \Omega_e = \frac{D}{D_e}\,\frac{V_e}{V}\,\Omega = \frac{1.5}{0.924} \times 1.0618 \times 2500 = 1.6229 \times 1.0618 \times 2500 = 4308 \text{ rpm}$$

The ratios $V/V_e, \Omega/\Omega_e$ are dimensionless, so km/h and rpm can be kept.

Power of the Airplane

With the three groups equal, $W/(\rho V^3 D^2)$ is equal:

$$\frac{W}{\rho V^3 D^2} = \frac{W_e}{\rho_e V_e^3 D_e^2} \implies W = \frac{\rho}{\rho_e}\left(\frac{V}{V_e}\right)^3 \left(\frac{D}{D_e}\right)^2 W_e$$

$$W = 0.6008 \times (0.9418)^3 \times (1.6229)^2 \times 6 \times 10^7 = 0.6008 \times 0.8354 \times 2.6338 \times 6 \times 10^7 = 7.93 \times 10^7 \text{ W}$$

Answers and Checks

- Reduced law: $W/(\rho V^3 D^2) = f\,(\rho V D/\mu,\ V/c,\ \Omega D/V)$.
- Model: $D_e = 0.924$ m, $V_e = 319$ km/h, $\Omega_e = 4308$ rpm.
- Airplane power: $W = 7.9 \times 10^7$ W.
- Dimensions: as in DA1, $\rho V^3 D^2$ is a power; V/c and $\Omega D/V$ are pure numbers.
- Why the model is smaller: sea-level air is denser ($\rho_e/\mu_e = 6.9 \times 10^4$ s/m^2 against 4.5×10^4 at 5000 m), so the same Reynolds number needs a smaller diameter; the faster tunnel speed (same Mach in warmer air) helps too.
- On the data: the given $W_e = 6 \times 10^7$ W is far above the real power of a small propeller (of order 10^5 W); the conversion factor $W/W_e = 1.32$ does not depend on it.
- Check against the official key: DA9.pdf gives $D_e = 0.924$ m, $V_e = 319$ km/h, $\Omega_e = 4308$ rpm and $W = 7.9 \times 10^7$ W, the same results.